Learning Statistically Accurate Dynamics of Complex Systems

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1 Previous Work

To understand the Earth system under a tractable computational lens, Behnoudfar et al. [1] proposed an explainable AI framework to systematically bridge the gap between coarse-grained conceptual models and high-resolution operational models. The idea is to integrate the physical interpretability and targeted statistical accuracy of idealized models in the forecast of black-box operational models. The proposed bridging framework consists of four components:

(I) A **physically-interpretable augmented latent space**, defined by an autoencoder that addresses the curse of dimensionality of high-dimensional data assimilation [2]. At each time t_k, the autoencoder compresses the normalized high-dimensional operational-model state (N physical variables over an $h \times w$ spatial grid) into a low-dimensional latent space (*encoding*), which, through the trained *decoder*, can be mapped back to the original state space. However, while standard autoencoders reduce state dimensionality, their latent spaces often lack direct physical interpretability, limiting their utility in integrating physics-based insights. To address this, the latent space is enriched with observed physical quantities, forming an augmented latent space. This ensures that the representation is both computationally efficient and physically grounded.

(II) A **data-driven short-term forecast model in the augmented latent space**, defined through a two-layer long short-term memory (LSTM) network [3]. This network is trained to emulate the short-term dynamical evolution of the operational model in the augmented latent space. Its output serves as the background *forecast* (*prior*) in the subsequent data assimilation cycle (see **(III)**), efficiently propagating the operational-model state forward so that it can be corrected using the idealized models' physical insights.

(III) A **model-bridging data assimilation** in the augmented latent space via an ensemble Kalman filter (EnKF) [4] (with covariance localization and inflation). This is the core "bridging" step. Here, we do not assimilate real-world observations, but instead systematically inject the targeted insights of idealized models, i.e., simulations from the conceptual models that play the role of *pseudo-observations* that are integrated with the operational LSTM forecast of the latent state. While data assimilation is traditionally regarded as a tool for state estimation, here it is repurposed as a mechanism for integrating model hierarchies. The pseudo-observations are not real measurements, but instead outputs generated by a cheap idealized model that provides targeted dynamical and statistical information. For this reason, the observation operator is accordingly modified to exploit the sparse yet physically meaningful output from coarse-grained idealized models (i.e., it only selects the coordinates of the augmented state vector corresponding to the pseudo-observations or physical observables),

thereby directly injecting their accuracy into operational surrogates. To achieve this operational bridge, the autoencoder from **(I)** is trained to ensure both: (i) *minimization of the error between the input and reconstructed state (produced by the decoder)*, and (ii) *maximization of the cross-correlation between the latent (encoded) states and the physical pseudo-observations*

(IV) A **curriculum learning method for structural bias correction**. A key challenge to building a skilful bridging model is to integrate not only the high-fidelity outputs from idealized models, but also limited reanalysis data. While the latent data assimilation corrects the forecast state using idealized models, the autoencoder and LSTM are initially trained on the biased operational model output, leaving certain structural errors uncorrected. To mitigate these biases, a curriculum learning strategy [5] is employed. By gradually blending limited reanalysis data (retrospective analyses of historical observations) into the training process, the foundational representation is corrected, resulting in a more accurate and observationally-constrained latent space for all subsequent steps.

This entire procedure facilitates the direct and stable assimilation of improvements from idealized models into operational models. It is inherently explainable, as the impact of each idealized model correction is traceable through the augmented latent space, providing a clear physical rationale for the resulting enhancements.

2 Motivation of Extension

In the baseline framework described in Section 1, the pseudo-observations provided by the idealized models are *pointwise* physical quantities (e.g., sea surface temperature and thermocline depth at individual grid points along the equator). The observation operator in the data assimilation step **(III)** is therefore a simple coordinate selector on the augmented state. Combined with the cross-correlation regularization in step **(I)**, this means that an EnKF correction applied to the pseudo-observation coordinates propagates to the encoded latent variables and, through the decoder, to the reconstructed high-resolution field.

However, in many practical settings, the available observations do not consist of pointwise measurements of the system state (direct or indirect), but the information is instead accessible in statistical, aggregated, or distributional form, such as such as spatial means, variances, covariances, or higher-order moments. Examples include climate diagnostics, coarse-grained measurements, ensemble summaries, long-time or phase-conditioned averages, and energy spectra or their dual, auto-/cross-correlations [6]. Such observations constrain the *distribution* of the state rather than a single realization. As a result, the inverse problem of reconstructing the state from such observations is inherently "ill-posed": *many distinct state configurations can produce the same finite collection of moments*, with the full task being inherently *infinite-dimensional*.

Tractable computational methods for solving this problem exist, based on finite-dimensional approximations (e.g. via Galerkin projection) or ensemble-/particle-based methods *on the state space* (e.g. the ensemble Fokker–Planck filter [6]). This project's objective is not to solve the infinite-dimensional problem directly, but instead to construct a tractable finite-dimensional approximation that preserves its core idea: assimilate statistical information rather than pointwise observations. We aim to achieve this by learning a low-dimensional latent representation of the high-dimensional state and by defining the statistical observation model in the latent space through decoded moments. In this framework, the dynamics regularize the otherwise ill-posed moment-inversion problem, while the autoencoder and decoder provide a learned closure that makes this regularization computationally usable.

3 Methodology

A schematic illustration of our proposed framework is shown in Figure 1.

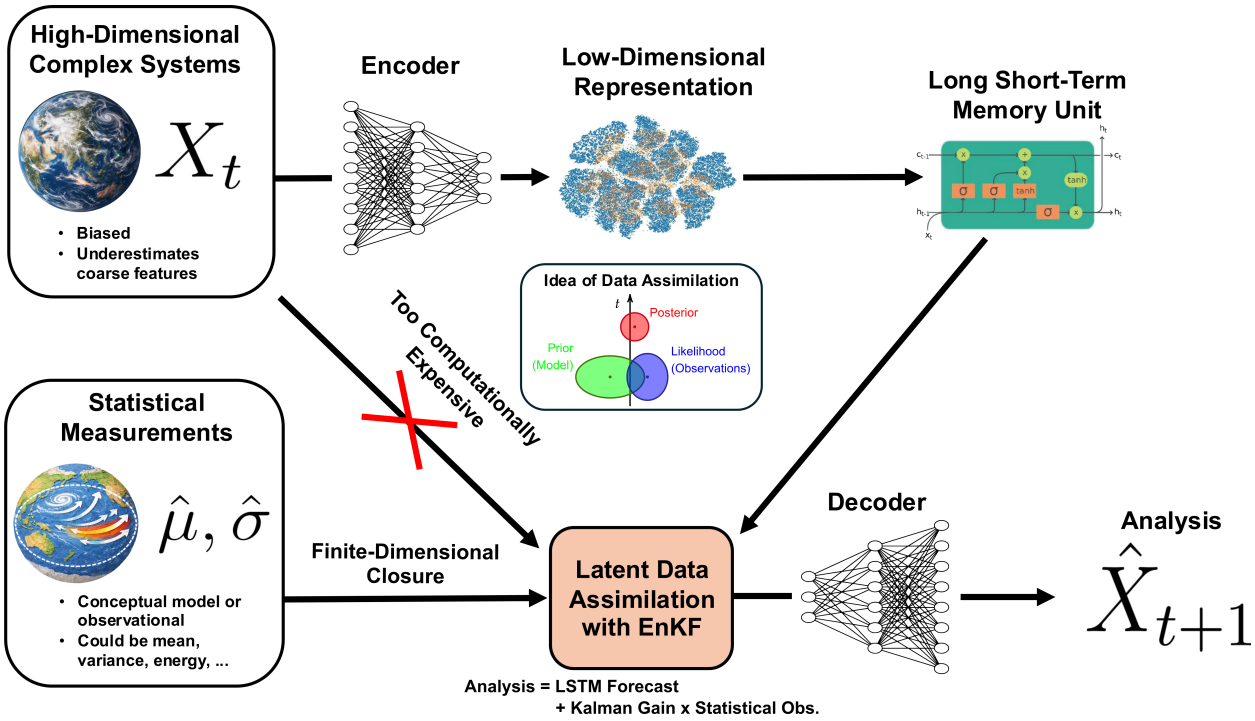


Figure 1: Schematic diagram of the extended framework with observable statistics.

3.1 Distributional Observations

Let $\mathbf{x}(\mathbf{s}, t) \in \mathbb{R}^N$ denote a spatially distributed state (discretized over an $h \times w$ grid) evolving in time (discretized over $K+1$ time steps). At discrete observation times $t_k = k\Delta t$, $k = 0, 1, \ldots, K$, we assume access to statistical moments of the state,

$$\mathbf{m}(t_k) = \boldsymbol{\mathcal{M}}\big(\mathbf{x}(\cdot, t_k)\big) + \boldsymbol{\varepsilon}_k, \tag{1}$$

where $\boldsymbol{\mathcal{M}}$ is a nonlinear operator that computes moments over space (e.g., mean, variance) or other statistical/coarse-grained quantities, and $\boldsymbol{\varepsilon}_k$ represents observation noise.

These observations are *distributional* rather than pointwise. The map $\boldsymbol{\mathcal{M}}$ is many-to-one (*multiple distinct states can yield the same moments*) and therefore non-invertible, meaning the state cannot be uniquely recovered from moment observations alone.

3.2 Latent Representation and Learned Closure

To regularize this ill-posed inverse problem, a learned latent representation is introduced, following the same dimensionality-reduction strategy as in step **(I)** of the baseline framework. An autoencoder is trained to map (in a physically meaningful way) the state to a low-dimensional latent variable, $\mathbf{E}(\mathbf{x})$, where $\mathbf{E}$ is the encoder, and to then reconstruct the full-dimensional state from the latent variable through a decoder $\mathbf{D}$, $\hat{\mathbf{x}} := \mathbf{D}(\mathbf{E}(\mathbf{x}))$.

As in the baseline framework, the latent variable $\mathbf{E}(\mathbf{x})$ (more broadly, the augmented state) is assumed to evolve according to a reduced-order dynamical model (through a learned LSTM network; see step **(II)**), enabling sequential forecasting and filtering in the latent space.

The decoder $\mathbf{D}$ implicitly defines a *closure*, i.e., a mapping from the latent variables to the full state realizations that "fills in" the unresolved degrees of freedom. When combined with the moment operator $\boldsymbol{\mathcal{M}}$, this closure provides a learned, data-driven relationship between statistical moments and plausible state configurations. In other words, even though many states share the same moments, the decoder selects a particular reconstruction consistent with the learned latent representation which, when coupled with the latent forecast model, yields trajectories compatible with the learned reduced dynamics.

3.3 Latent-Space Filtering with Moment Observations

This step is the analogue of the model-bridging data assimilation procedure in step **(III)** of the baseline framework, but with a different observation operator. In the baseline, the observation operator $\mathbf{h}$ simply selects the pseudo-observation coordinates from the augmented state vector (linear, physical observables coordinate-selector in the augmented state $\mathbf{a} = \big(\mathbf{x}, \mathbf{z}\big)^{\mathsf{T}}$; $\mathbf{h} = \begin{pmatrix}\mathbf{0} & \mathbf{I}\end{pmatrix}$). Here, the observation operator is fundamentally different, being the composition $\boldsymbol{\mathcal{M}} \circ \mathbf{D}$: first decode the latent variable into a full state, then compute its moments. Therefore, for the encoded state $\mathbf{E}(\mathbf{x}(t_k))$, the reconstructed moments are

$$\hat{\mathbf{m}}(t_k) = \boldsymbol{\mathcal{M}}(\mathbf{D}(\mathbf{E}(\mathbf{x}(t_k)))) = \boldsymbol{\mathcal{M}}(\hat{\mathbf{x}}(t_k)) = \mathbf{h}(\mathbf{E}(\mathbf{x}(t_k))),$$

and filtering is performed in the latent space by defining a likelihood (e.g., Gaussian) based on these reconstructed moments and (1) by

$$p\big(\mathbf{m}(t_k) \mid \mathbf{E}(\mathbf{x}(t_k))\big) \propto \exp\Big(-\tfrac{1}{2}\big\|\mathbf{m}(t_k) - \mathbf{h}(\mathbf{E}(\mathbf{x}(t_k)))\big\|^2_{\mathbf{R}_k^{-1}}\Big), \tag{2}$$

where $\mathbf{R}$ is the moment-observation error covariance, $\boldsymbol{\varepsilon}_k \overset{\mathrm{d}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$. Sequential Bayesian updates (e.g., via the EnKF) are then applied to the latent variables using this likelihood, with the the learned LSTM dynamics acting as the forecast prior. Importantly, the observation operator acts only through statistical moments, not through direct state observations.

As in the baseline, for the assimilation to correct all latent components (not just those directly related to the observed moments), statistical dependence between the latent representation and the moments must hold.

3.4 Coupling Latent Variables and Moments with Cross-Covariance Regularization

This step is analogous to the augmented latent space in step **(I)** of the baseline framework, where the latent vector was concatenated with pseudo-observations from the idealized model, the latent variable is decomposed as (using subscripts $\cdot_k$ to denote calculation at time t_k for brevity):

$$\mathbf{a}_k = \begin{pmatrix}\mathbf{E}(\mathbf{x}_k) \\ \mathbf{m}_k\end{pmatrix}, \tag{3}$$

where $\mathbf{E}(\mathbf{x}_k) \in \mathbb{R}^{n_l}$ encodes geometric or dynamical structure as the output of the encoder and $\mathbf{m}_k \in \mathbb{R}^{n_m}$ contains the statistical moments of the state in physical space (e.g., the spatial mean and variance of each field).

To encourage information transfer during assimilation, *just as in the baseline framework*, a regularization term is introduced that promotes non-zero statistical dependence between $\mathbf{E}(\mathbf{x}_k)$ and $\mathbf{m}_k$:

$$\mathcal{L}_{\text{corr},k} = -\lambda \left\| \text{Corr}(\mathbf{E}(\mathbf{x}_k), \mathbf{m}_k) \right\|_{\text{F}}^2 . \tag{4}$$

$\| \cdot \|_{\text{F}}$ denotes the Frobenius norm and the negative sign means that while maximizing this term, thus encouraging a large cross-correlation between the latent variables and the moments, we are minimizing the composite autoencoder loss during training. This is the direct analogue of the cross-correlation loss $\mathcal{L}_{\text{corr}}$ in the baseline framework. Through this, when the EnKF computes the Kalman gain, the non-zero off-diagonal blocks of the ensemble covariance (between $\mathbf{E}(\mathbf{x}_k)$ and $\mathbf{m}_k$) allow corrections applied to the moment components to propagate into the encoded latent variables, without enforcing deterministic equality or collapsing the learned representation.

The rationale is the fact that the Kalman gain is proportional to the (forecast) cross-covariance between the forecast state and predicted (statistical) observations; if $\text{Cov}(\mathbf{E}(\mathbf{x}_k), \mathbf{m}_k)$ is close to zero, assimilating $\mathbf{m}_k$ will barely change $\mathbf{E}(\mathbf{x}_k)$ thus decreasing the data assimilation skill significantly.

4 Results

We applied this framework to two different models.

4.1 Stochastic Lorenz '96 Model

Our framework is first applied to the stochastic Lorenz '96 (L96) model, given by the stochastic differential equation (SDE):

$$dx_i = (x_{i-1}(x_{i+1} - x_{i-2}) - x_i + F)dt + \sigma dW_i(t), \quad \text{for } i \in \{1, 2, \ldots, N\},$$

where the indices follow periodic boundary conditions, i.e. $x_0 = x_{40}$, $x_{-1} = x_{39}$. We used the constants $N = 40$, $F = 8$, $\sigma = 0.5$. L96 loosely models the energy of our atmosphere, where the $x_{i-1}(x_{i+1} - x_{i-2})$ term represents advection towards higher index components, the $-x_i$ term represents damping, and F is a forcing term.

We mainly considered an observation operator that would measure the true value of a small subset of the components, x_i. For example, imagine there are a few weather observatories on the area modeled by the L96 model that measure the local energy in the atmosphere. More formally, for a subset $S \subset \{1, 2, ..., N\}$, we define $\mathcal{M}_S(X) = \{(x_i, i) : i \in S\}$.

Some of the results that we achieved are shown in Figures 2–3.

4.2 Stochastically Coupled FitzHugh-Nagumo Model

The second model we applied our framework to is the stochastic FitzHugh-Nagumo (FHN) model. Specifically we used a locally coupled version of the model. It is given by the following coupled SDEs:

$$\varepsilon du_i = (d_u(u_{i-1} - 2u_i + u_{i+1}) + u_i - cu_i^3 - c_1 v_i)dt + \sqrt{\varepsilon}\delta_1 dW_{u,t}(t),$$

$$dv_i = (u_i + a - c_2 v_i)dt + \delta_2 dW_{v,i}(t), \quad i = 1, 2, \ldots, N.$$

These use the periodic boundary conditions $u_N = u_0, u_{N+1} = u_1$. We use the constants $c = 1/3, a = 1.05, \varepsilon = 10^{-2}, c_1 = 1, c_2 = 0, \delta_1 = 0.2, \delta_2 = 0.4, d_u = 0.5, N = 40$.

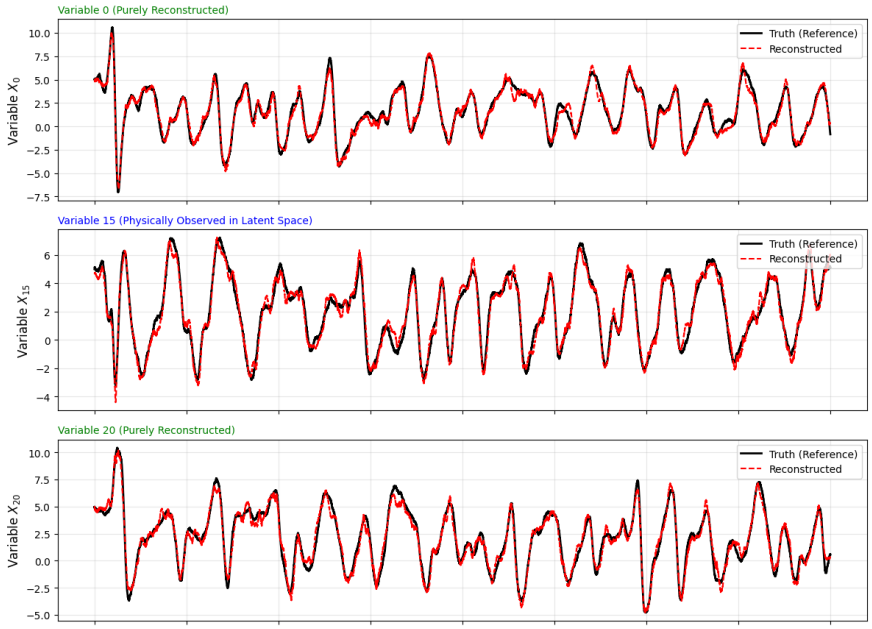


Figure 2: Results of the autoencoder used for the L96 model. Red is the predicted values and black is the ground truth. The middle row was one of the few components observed by our observation operator, while the outer rows were not.

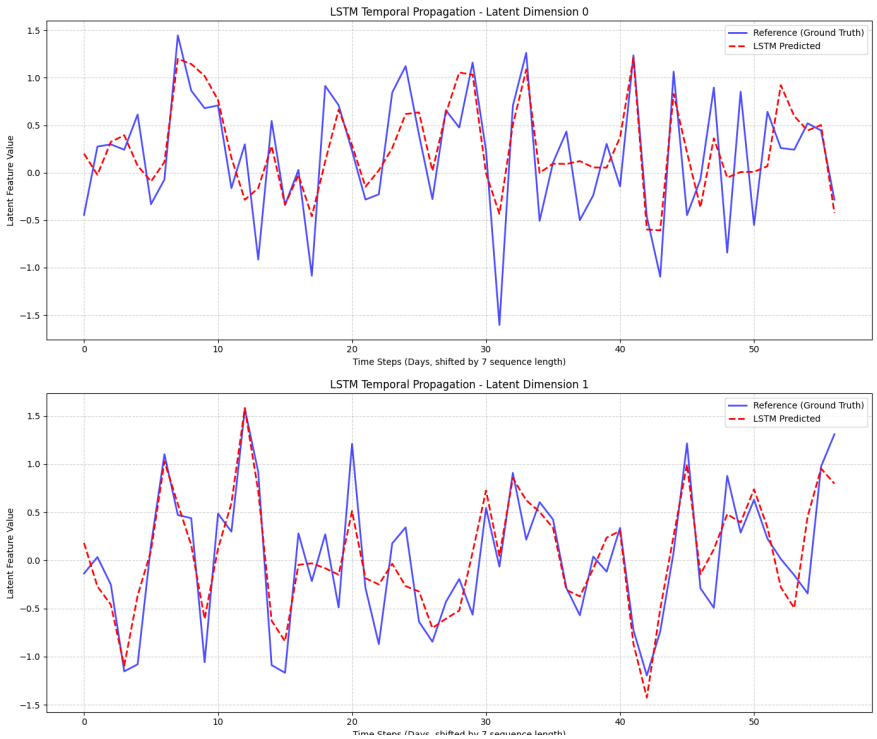


Figure 3: Results of the long-short term memory unit used to model the dynamics in the latent space. The red is the predicted values while the blue is the ground truth values, computed using the autoencoder.

This system simulates a simplified neuronal dynamical system, capturing the dynamics of action potentials. The time scale separation ratio, $\varepsilon \ll 1$, creates the fast–slow dynamics characteristic of neuronal spiking: The membrane voltage u changes rapidly during an action potential, while the recovery variable v evolves on a slower time scale.

We considered several moment observation operators. Firstly, we consider the means, variances and covariance of the system:

$$\overline{u} = \frac{1}{N}\sum_{n=1}^{N} u_n, \quad \overline{v} = \frac{1}{N}\sum_{n=1}^{N} v_n, \quad V_u = \frac{1}{N}\sum_{n=1}^{N}(u_n - \overline{u})^2, \quad V_v = \frac{1}{N}\sum_{n=1}^{N}(u_n - \overline{u})^2, \quad C_{uv} = \frac{1}{N}\sum_{n=1}^{N}(u_n - \overline{u})(v_n - \overline{v}).$$

Based on these moments, we define the following operators which are concatenated and used in our implementation:

$$\mathcal{M}_{spike}(u,v) = \left(\frac{1}{N}\sum_{n=1}^{N}\mathbf{1}_{\{u_n>\theta\}}, \overline{u}, V_u\right), \quad \mathcal{M}_{corr}(u,v) = \frac{C_{uv}}{\sqrt{V_u + \varepsilon_{uv}}\sqrt{V_v + \varepsilon_{uv}}},$$

where $\varepsilon_{uv} = 10^{-6}$. We perform data assimilation with these operators to ensure the spiking behavior of the system can be recreated. After applying the framework, we got the following results shown in Figures 4–7.

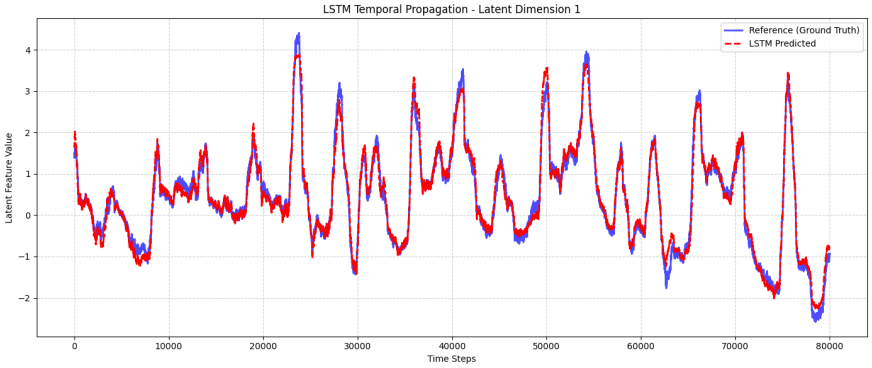


Figure 4: Results of the LSTM modeled dynamics in the latent space.

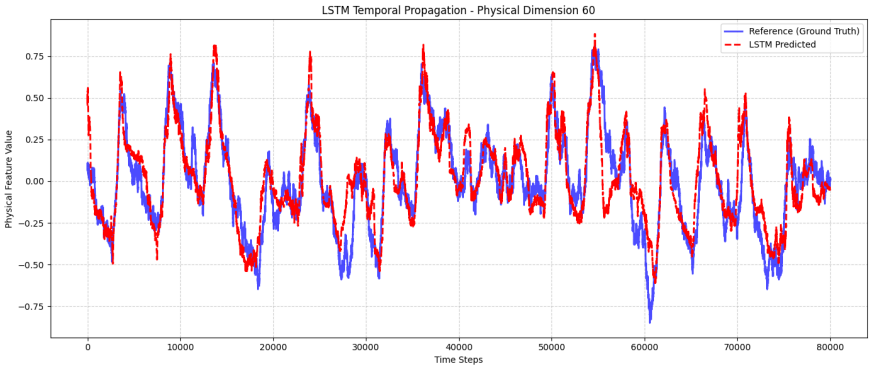


Figure 5: Ground truth of the system versus predictions from LSTM in latent space after decoding. Time series for one dimension of the v (slow) variable.

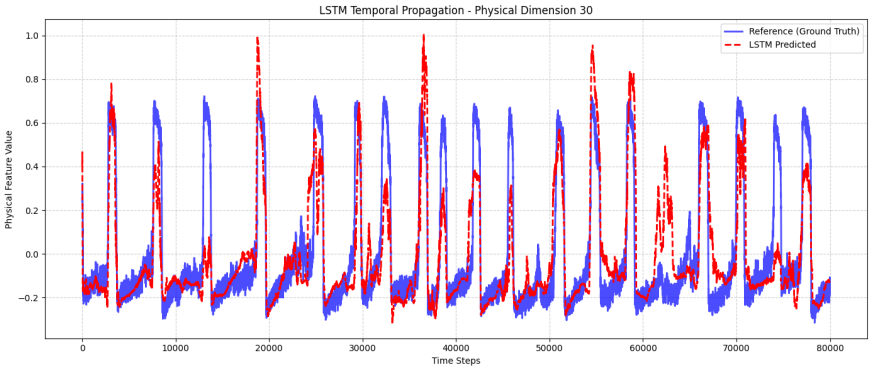


Figure 6: Ground truth of the system versus predictions from LSTM in latent space after decoding. Time series for one dimension of the u (fast) variable.

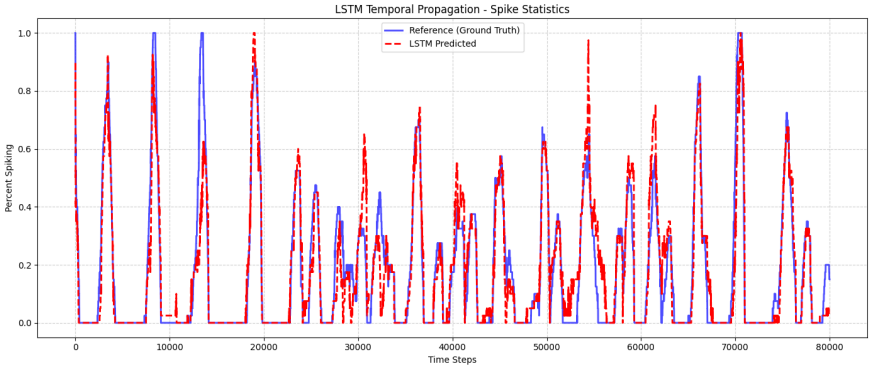


Figure 7: The first dimension of $\mathcal{M}_{spike}$ evaluated on the training data versus on the predictions from the LSTM after decoding.

From the u time series and the percentage spiking observation, we can see that applying our framework led to an accurate recreation of spike behavior, as desired.

5 Discussion

Our cross-covariance-regularization approach ensures moment-based observations constrain the latent state despite the absence of direct state measurements. By providing explicit control over statistical coupling it is well-suited for ensemble-based filters and enables data assimilation with distributional observations through a learned closure, providing a flexible framework for systems where only statistical information is observable (or where accurate statistical measurements can be provided from conceptual models).

We should note that alternative approaches have been tested so as to couple the latent variables and statistical moments for data assimilation; instead of explicit augmentation, moments can be modeled as functions of the latent variables where the observation operator for latent data assimilation is nonlinear and modeled via a small auxiliary neural network instead. The new training objective in this regime includes a moment reconstruction loss to enforce statistical moments as implicit properties of the latent representation. One noticeable advantage of this approach is that the forecast model and EnKF run strictly in the smaller latent space, but the observation operator

becomes nonlinear, since each ensemble member must be pushed through the nonlinear network, and entangled with the encoder, thus requiring careful employment considerations. A two-hidden-layer MLP is usually a sufficient architecture for simple models with instantaneous statistics (e.g. the L96 model with block averages or low-order moments); however, for excitable fast-slow systems such as FitzHugh-Nagumo, a more complex architecture that is channel-aware and considers lagged or temporal statistics in its training regime is required to be robust against spikes or synchronized episodes.

6 Summary

The proposed extension performs sequential Bayesian filtering (e.g., via the EnKF) in a learned latent space, where observations consist of statistical moments of the state rather than pointwise measurements ("pseudo-observations"). A data-driven closure, learned via an autoencoder as in step **(I)** of the baseline framework, provides the inverse mapping from moments to plausible state distributions. Cross-covariance regularizon ensures that moment-based corrections propagate into the full latent representation, generalizing the cross-correlation mechanism that defines the baseline's model-bridging data assimilation in step **(III)**.

Notably, the assimilation of distributional observations does not aim to recover individual trajectories in a chaotic system, but rather to constrain the evolving ensemble toward the correct invariant measure or statistical attractor, i.e. the long-run statistical distribution of the system's states (although for some systems, if moment observations are sufficiently informative and frequent, some trajectory-level correction is still possible).

7 Acknowledgement

We would like to thank our project leader **Pouria Behnoudfar** and our graduate mentor **Marios Andreou** for their advice throughout the project.

8 References

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